

Interface-Enhanced Superconductivity in Ultrathin TiN Proximitized by Topological Insulators

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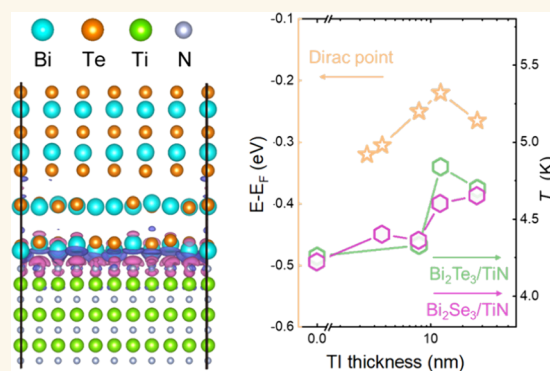
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ABSTRACT: A high-quality topological insulator–superconductor (TI–SC) heterostructure with an atomically sharp and well-controlled interface is crucial for realizing topological superconductivity and a topological quantum qubit. In particular, many studies of TI–SC heterostructures have focused on inducing a superconducting gap in the TI layer via proximity effect, while the active manipulation of superconductivity in the SC layer remains largely unexplored. In this work, we fabricated TI/TiN heterostructures using highly air-stable, ultrathin TiN films as the SC layer and observed an interface-enhanced superconductivity that contrasts with the conventional proximity effect in the superconductor–normal metal interface. Band structure measurements reveal a consistent shift of the Dirac point with T_c enhancement. Interfacial charge transfer provides a plausible explanation for this shift based on the systematic analysis and is therefore a likely contributor to the observed T_c enhancement. First-principles calculations elucidate the charge transfer pathways, highlighting the critical role of the interfacial BiTe (BiSe) bilayer. Our results not only provide a tunable TI–SC hybrid system with robust superconductivity at ultrathin thickness but also offer a potential route for manipulating superconductivity in TI–SC heterostructures via interface engineering.

KEYWORDS: topological insulator–superconductor, interface-enhanced superconductivity, charge transfer, ultrathin TiN, interface engineering



Superconductivity is a macroscopic quantum phenomenon characterized by zero electrical resistance below a critical temperature (T_c). Integrating superconductivity with topological materials can give rise to a topological superconductor, an exotic form of quantum matter predicted to host Majorana zero modes (MZMs). These quasiparticles, which obey non-Abelian statistics, are considered to be building blocks for realizing fault-tolerant quantum computing. An effective approach to implement topological superconductors involves coupling an s-wave superconductor with a topological insulator, forming a topological insulator–superconductor (TI–SC) heterostructure.^{1–3} Various theoretical proposals for realizing MZMs, including those based on vortex cores at the TI–SC interface or edge states in quantum anomalous Hall insulator–superconductor heterostructures,^{4–6} require the growth of high-quality TI–SC heterostructures with an atomically sharp and well-controlled interface.

Among various TI–SC heterostructures,^{7–17} the ones utilizing 3D TIs like Bi_2Te_3 or Bi_2Se_3 outperform other materials due to their spin nondegenerate helical surface states and a large band gap of ~ 0.3 eV.¹³ Compared to the readily obtained high-quality TI layer facilitated by van der Waals

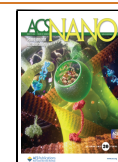
epitaxy, achieving a high-quality SC layer remains the major challenge in growing TI–SC heterostructures. First, most suitable superconducting materials are chemically reactive,^{7,10,18} and some of them tend to react with the TI layer, resulting in interfacial degradation or the formation of a dead layer.⁷ This high reactivity also renders the SC layer air-sensitive and prone to surface degradation, requiring stringent *in situ* growth^{10,12,13,19–21} or *in situ* cleaving processes.^{8,9,22–24} Furthermore, the superconductivity in most superconducting thin films deteriorates rapidly as the thickness is reduced to the atomic scale, often leading to a broadened superconducting transition or even complete loss of superconductivity.^{13,19,25,26} Due to the lack of ultrathin SC films with sharp transition, prevailing studies focus on inducing superconducting gaps in

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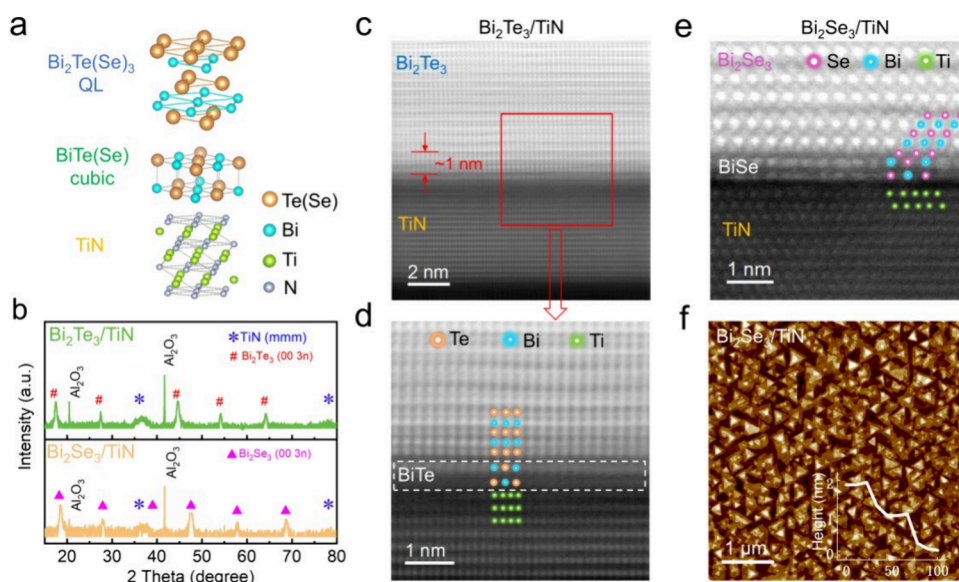


Figure 1. Structural characterizations of $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)/\text{TiN}$ heterostructures. (a) Schematic illustration of the $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)/\text{TiN}$ heterostructure. (b) XRD profiles of 25 nm $\text{Bi}_2\text{Te}_3/\text{TiN}/\text{Al}_2\text{O}_3$ and 50 nm $\text{Bi}_2\text{Se}_3/\text{TiN}/\text{Al}_2\text{O}_3$. The marks labeled $\text{Bi}_2\text{Te}_3(00\ 3n)$, $\text{Bi}_2\text{Se}_3(00\ 3n)$, and TiN peaks. (c) STEM-HAADF image of the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure. The red arrows indicate that the thickness of 1 QL is about 1 nm. (d) An enlarged image of (c), a cubic BiTe bilayer, is clearly visible at the interface. (e) STEM-HAADF image of the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure. A similar BiSe bilayer is also observed at the interface. (f) AFM morphology of a 25 nm $\text{Bi}_2\text{Se}_3/\text{TiN}/\text{Al}_2\text{O}_3$ sample. The inset shows the height profile of the terraces with a step height of ~ 1 nm, corresponding to 1 QL of Bi_2Se_3 .

the TI layer,^{7,8,24,27} while the reverse effect, i.e., how to manipulate the superconductivity in the SC layer via proximity coupling with the TI layer, remains largely unexplored. The shunting effect from the thick SC layer renders the subtle interactions at the TI-SC interface very challenging to detect by transport measurements, obscuring the understanding of interactions between the SC and TI layers at the atomically thin interface.

In this work, we introduce an *ex situ* grown ultrathin TiN(111) film as the desired SC layer that demonstrates extraordinary environmental stability.^{28–31} The 6-fold structural symmetry of TiN(111) enables high compatibility with TIs, resulting in superior interfacial quality. Compared to superconductors like $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$,³² $\text{Nd}_{0.8}\text{Sr}_{0.2}\text{NiO}_2$,³³ NbSe_2 ,³⁴ and FeSe ,³⁵ which are prone to degradation in air, however, the robust TiN surface is resistant to degradation, contamination, and even acid or alkali corrosion.^{36–39} We used a magnetron sputtering epitaxy system to grow ultrathin TiN films as SC layers and transfer them to a molecular beam epitaxy (MBE) chamber to fabricate TI/TiN heterostructures. We demonstrate that the TiN films preserve robust superconductivity with sharp transitions at an ultrathin thickness of ~ 4 nm, thereby minimizing bulk effects and enabling clear probing of variation in superconductivity. Most surprisingly, we observed an enhancement of superconducting temperature T_c in $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)/\text{TiN}$ heterostructures compared to a pure TiN film. This phenomenon contrasts sharply with the conventional superconducting proximity effect at a superconductor–normal metal interface, where the proximity coupling quenches the superconductivity and lead to a T_c reduction in the SC layer.^{40,41} The origin of the enhanced superconductivity is investigated by transport measurements, angle-resolved photoemission spectroscopy (ARPES), and first-principles calculations, providing valuable insights on interfacial manipulation of superconductivity in the TI-SC system.

RESULTS

Structural Characterizations of the TI/TiN Heterostructures

Figure 1a depicts a schematic illustration of the TI (Bi_2Te_3 or Bi_2Se_3)/TiN heterostructures. Notably, a cubic BiTe (BiSe) bilayer naturally formed at the interface between TI and TiN layers during the growth, which is characterized by scanning transmission electron microscopy (STEM) imaging in Figure 1d,e. Bi_2Te_3 and Bi_2Se_3 are benchmark 3D TIs with a single Dirac cone located at the Γ point of the first Brillouin zone.⁴² Their unit cells are composed of quintuple-layer (QL) blocks with a Te(Se)–Bi–Te(Se)–Bi–Te(Se) atomic arrangement. Figure 1b gives the X-ray diffraction (XRD) 2θ scans for both the $\text{Bi}_2\text{Te}_3/\text{TiN}$ and $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures, which clearly show the $\text{Bi}_2\text{Te}_3(00\ 3n)$, $\text{Bi}_2\text{Se}_3(00\ 3n)$, and TiN(*mmm*) peaks, without any sign of secondary phases. The rocking curve scans in Figure S1 yield full width at half-maximum (fwhm) values of approximately 0.1° for the $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure and 0.06° for the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure. The XRD results together with the distinct low-energy electron diffraction (LEED) patterns (Figure S4) indicate the high-quality epitaxy for TI/TiN heterostructures, presumably owing to the same 6-fold symmetry between TiN(111) and Bi_2Te_3 (Bi_2Se_3).³⁶

Heterostructures grown on *ex situ* films typically exhibit degraded surface quality or even interfacial dead layers. However, the STEM images (Figure 1c–e and Figure S5) of both $\text{Bi}_2\text{Te}_3/\text{TiN}$ and $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures grown on *ex situ* TiN films (as thin as 4.2 ± 0.2 nm) demonstrate atomically sharp and well-defined interfaces. Figure 1d shows an enlarged view of the $\text{Bi}_2\text{Te}_3/\text{TiN}$ interface, revealing a distinct bilayer structure (~ 3.3 Å thick) that differs from that of the Bi_2Te_3 phase. Based on a careful analysis of the atomic spacing (Figure S6), we identify this bilayer structure as the cubic BiTe phase. A similar cubic BiSe bilayer is also observed at the $\text{Bi}_2\text{Se}_3/\text{TiN}$ interface (Figure 1e) and has previously

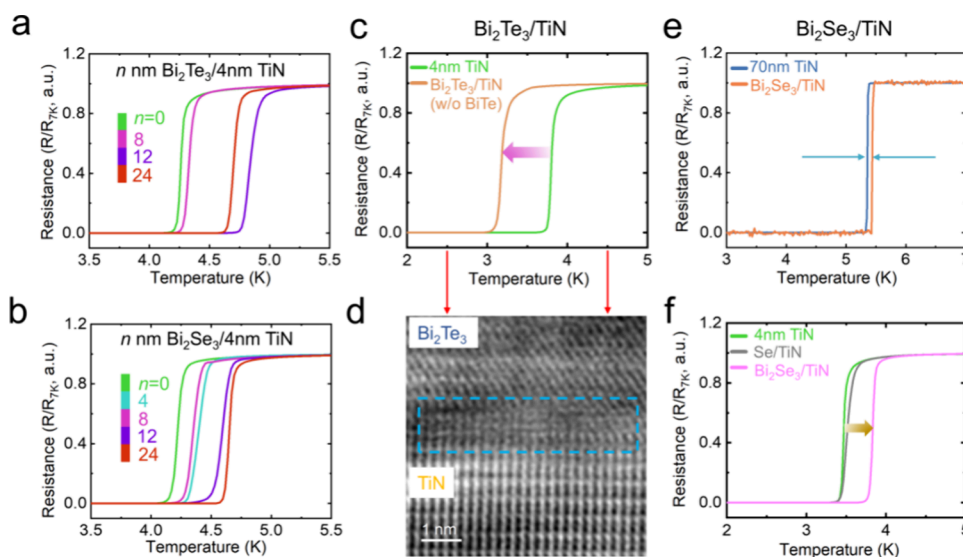


Figure 2. Superconductivity enhancement in $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)/\text{TiN}$ heterostructures. R – T curves of (a) n nm $\text{Bi}_2\text{Te}_3/4$ nm TiN and (b) n nm $\text{Bi}_2\text{Se}_3/4$ nm TiN heterostructures from 3.5 to 5.5 K under a zero magnetic field. (c) R – T curves of 4 nm TiN and 12 nm $\text{Bi}_2\text{Te}_3/4$ nm TiN with a deteriorated interface. The $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure shows a T_c reduction of 0.6 K compared with pure TiN . (d) STEM-HAADF image of the $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure in (c). The blue dashed box shows a deteriorated interface without BiTe bilayer intercalation. (e) R – T curves of 70 nm TiN and 50 nm $\text{Bi}_2\text{Se}_3/70$ nm TiN . The T_c of the two samples is nearly identical (change <0.1 K). (f) R – T curves of 4 nm TiN , 140 nm $\text{Se}/4$ nm TiN , and 50 nm $\text{Bi}_2\text{Se}_3/4$ nm TiN . The T_c of Se/TiN is the same as that of a pure TiN film, while the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure shows an enhancement of 0.4 K. In all the above images, R is normalized by the normal-state resistance at $T = 7.0$ K. All samples were grown on Al_2O_3 substrates except for the films in Figure 2b, which were grown on KTO_3 substrates.

been reported at the $\text{Bi}_2\text{Se}_3/\text{NbSe}_2$ interface.¹² The atomic force microscopy (AFM) morphology of the TiN film (Figure S8) exhibits a surface roughness comparable to that of commercial Al_2O_3 substrates³⁶ (146 pm), providing an atomically flat surface for the epitaxial growth of TI layers. The AFM images of both $\text{Bi}_2\text{Te}_3/\text{TiN}$ (Figure S8) and $\text{Bi}_2\text{Se}_3/\text{TiN}$ (Figure 1f) show characteristic triangular-shaped terraces with a height of ~ 1 nm, corresponding to 1 QL of Bi_2Te_3 or Bi_2Se_3 . These large, flat terraces suggest a good structural match between the TI layers and TiN surface,³⁶ which is in agreement with the XRD and STEM results.

Enhanced Superconductivity in TI/TiN Heterostructures

We performed *ex situ* electrical transport measurements on both pure TiN films and TI/ TiN heterostructures to evaluate their superconducting properties. We note that the superconducting transition temperature T_c of TiN films varies among different sputtering batches. Therefore, all data presented within a single figure are grown on TiN films from the same sputtering batch to ensure a rigorous comparison. Figure 2a compares the temperature-dependent normalized longitudinal resistance of a pure TiN film and $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures with varying thicknesses of Bi_2Te_3 (8, 12, and 24 nm), all fabricated on 4 nm TiN films from the same batch. The 4 nm ultrathin TiN film exhibits a sharp superconducting transition with a T_c (defined as the temperature at which R drops to 50% of the normal-state resistance measured at 7.0 K) of 4.3 K, demonstrating no degradation of superconductivity even at the near-atomic scale. After the deposition of Bi_2Te_3 on the TiN surface, a noticeable enhancement in T_c is observed for the $n = 8$ heterostructure, and the enhancement becomes more pronounced in the $n = 12$ and 24 heterostructures. The $n = 12$ sample reaches a maximal T_c of 4.9 K, about 0.6 K higher than that of pure TiN . In contrast, Figure 2c shows a 12 nm $\text{Bi}_2\text{Te}_3/4$ nm TiN control sample with a deliberately deteriorated interface lacking the BiTe bilayer, which exhibits

a T_c reduction of 0.6 K (Figure 2d). This result unambiguously validates that a high-quality interface with a BiTe bilayer is critical for the superconductivity enhancement.

Then, we replaced Bi_2Te_3 with Bi_2Se_3 and grew $\text{Bi}_2\text{Se}_3/\text{TiN}$ (Figure 2b) heterostructures, in which the T_c of pure TiN was measured as 4.2 K. The T_c of the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures increased gradually from $n = 4$ to $n = 24$, reaching a maximal value in the $n = 24$ sample with an enhancement of ~ 0.5 K compared to pure TiN . As a comparison, Figure 2e shows the resistance of a $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure grown on thick TiN (70 nm), exhibiting a nearly identical T_c (change <0.1 K) with pure TiN . Notably, STEM measurements reveal that the cubic BiSe interfacial phase is absent in this sample (Figure S7), in contrast to the $\text{Bi}_2\text{Se}_3/4$ nm TiN case. This structural difference is likely due to the rougher TiN surface with higher nucleation sites in thick TiN (Figure S8), which favors direct van der Waals Bi_2Se_3 nucleation over the metastable BiSe phase. Consequently, this structural difference provides a plausible explanation for the absence of T_c enhancement and likely reflects the distinct interfacial conditions in the thick TiN films. This observation implies that a smoother TiN surface with lower nucleation density may be a necessary condition for BiSe interfacial layer formation. Next, we replaced the TI layer with a nontopological material, Se , as shown in Figure 2f. The resistance curve of the 140 nm $\text{Se}/4$ nm TiN sample closely resembles that of the pure TiN film, while the 50 nm $\text{Bi}_2\text{Se}_3/4$ nm TiN heterostructure shows a clear T_c enhancement of 0.4 K, indicating that the TI layer could play an essential role in enhancing superconductivity.

Thickness-Dependent Evolution of the Band Structure

Next, we performed *in situ* ARPES measurements on $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures with varying Bi_2Te_3 thickness. The band structure of the ultrathin pure TiN film is difficult to resolve due to the requirement of *ex situ* transfer, considering that ARPES is a surface-sensitive probe. Figure S9 gives the

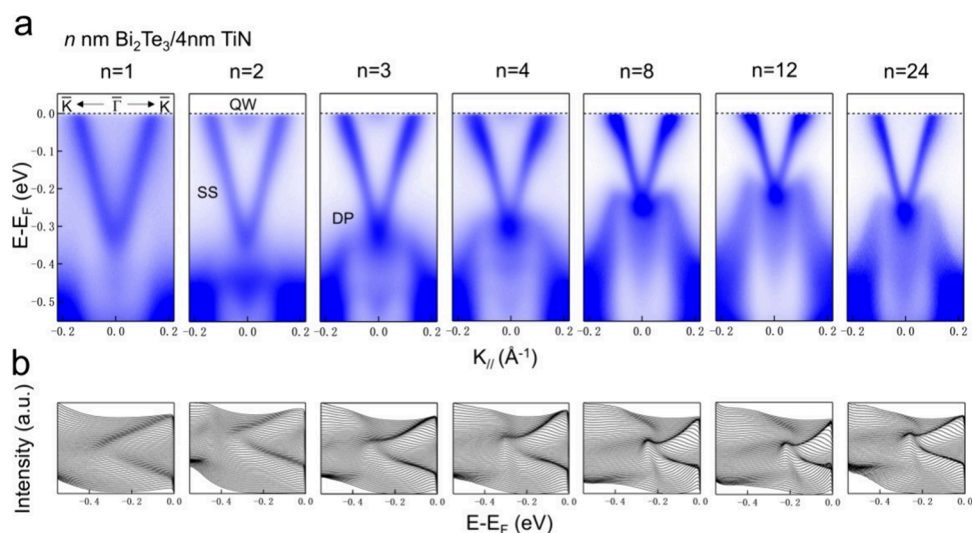


Figure 3. ARPES band spectra of n nm Bi_2Te_3 /4 nm TiN heterostructures. (a) ARPES results with $n = 1, 2, 3, 4, 8, 12$, and 24 . (b) Corresponding energy distribution curves of ARPES results in (a). All spectra were collected along the $\bar{\Gamma}$ – $\bar{K}$ direction. The bulk quantum well appears at $n = 2$ and disappears at $n = 8$. The surface state appears at $n \geq 2$, and the Dirac point is formed at $n = 3$. All the data were collected at 7.3 K.

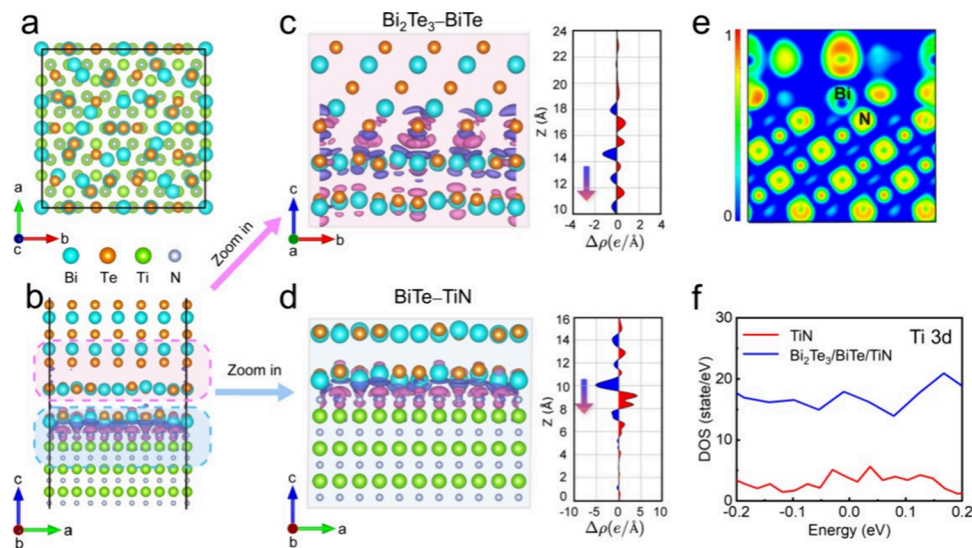


Figure 4. Theoretical calculations on $\text{Bi}_2\text{Te}_3/\text{BiTe}/\text{TiN}$ heterostructure. (a, b) Top and side views of the $\text{Bi}_2\text{Te}_3/\text{BiTe}/\text{TiN}$ heterostructure. (c, d) Differential charge-density distributions at the Bi_2Te_3 – BiTe (isosurface level = $0.002 \text{ e}/\text{\AA}$) and BiTe – TiN (isosurface level = $0.009 \text{ e}/\text{\AA}$) interfaces (left), along with the corresponding integrals of the differential charge densities (right). The positive isosurface is indicated in red, and the negative counterpart in blue, at a fixed isosurface level. (e) Electron localization function (ELF) at the BiTe – TiN interface. (f) Ti atom-resolved partial density of states (DOS) of TiN and the $\text{Bi}_2\text{Te}_3/\text{BiTe}/\text{TiN}$ heterostructure.

hexagonal Fermi surface mapping of a 50 nm Bi_2Te_3 /4 nm TiN heterostructure, which is consistent with a previous study on Bi_2Te_3 .⁴³ Figure 3 presents the ARPES spectra and corresponding energy distribution curves collected along the $\bar{\Gamma}$ – $\bar{K}$ direction for Bi_2Te_3 /4 nm TiN heterostructures with Bi_2Te_3 thickness ranging from 1 to 24 nm. For the 1 nm Bi_2Te_3 /TiN sample ($n = 1$), the ARPES spectra exhibit a nearly parabolic band dispersion, similar to that of 1 QL Bi_2Te_3 grown on the $\text{Si}(111)$ substrate.⁴⁴ In the $n = 2$ sample, a quantum well (QW) state appears in the bulk conduction band, indicating that the film thickness is within the quantum confinement regime. As the thickness increases, surface states form a gapless Dirac cone, causing the quantum well states to gradually disappear, overshadowed by the spectral weight of the topological surface states. Additionally, a new valence band emerges at $E - E_F \approx -0.45 \text{ eV}$ in the $n = 2$ sample, resulting in

a gap of approximately 0.13 eV to the conduction band. As the thickness of the Bi_2Te_3 layer increases, these two bands evolve into a gapless surface state, forming a typical Dirac cone. Meanwhile, the conduction and valence bands cross at $E - E_F \approx -0.32 \text{ eV}$, with the crossover thickness determined to be 3 nm, consistent with a previous study.⁴⁴ For samples with $n \geq 3$, the Dirac point gradually shifts toward the Fermi level with increasing thickness until it reaches a saturation value of approximately 24 nm. Although ARPES is surface-sensitive and does not directly probe the Bi_2Te_3 /TiN interface, it detects interface-induced modifications of the surface electronic structure. Interfacial charge transfer alters the surface chemical potential via band bending and charge redistribution, manifested as a shift of the Dirac point at the surface. Figure 5a summarizes the thickness-dependent shift of the Dirac point in Bi_2Te_3 /TiN alongside the corresponding T_c enhancement in

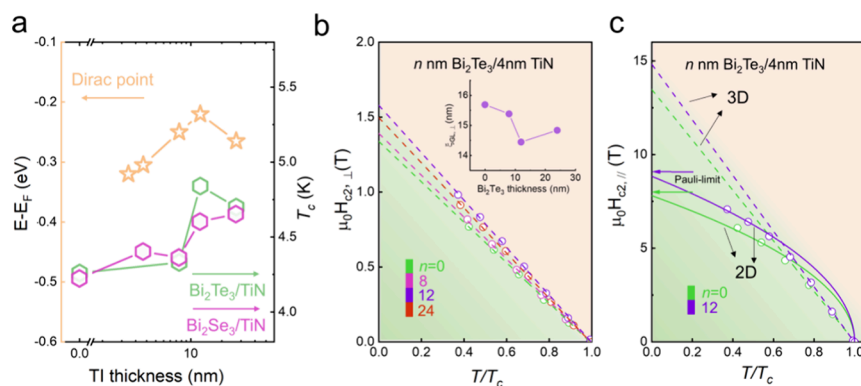


Figure 5. T_c and upper critical fields of $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)/\text{TiN}$ heterostructures. (a) Thickness dependence of the Dirac point ($\text{Bi}_2\text{Te}_3 \geq 3$ nm) and the T_c of $\text{Bi}_2\text{Te}_3/\text{TiN}$ ($\text{Bi}_2\text{Se}_3/\text{TiN}$) heterostructures. Upper critical fields ($H_{c2,\perp}$ and $H_{c2,\parallel}$) of the $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures with magnetic fields applied perpendicular (b) and parallel (c) to the ab plane. The inset in (b) shows the out-of-plane GL coherence lengths as a function of Bi_2Te_3 thickness. The dashed and solid lines in (c) represent fitting results by 3D and 2D GL theory, respectively. The green and purple arrows indicate the Pauli limit of $H_{c2,\parallel}$ for TiN and $\text{Bi}_2\text{Te}_3/\text{TiN}$, which are 8.0 and 9.1 T, respectively.

both $\text{Bi}_2\text{Te}_3/\text{TiN}$ and $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures. While the Dirac point in TI films can evolve with thickness due to intrinsic vacancies and antisite defects, control experiments on Bi_2Te_3 films grown on InP (Figures S10 and S11) under identical conditions show a much smaller maximum shift (~ 60 meV) than the ~ 100 meV shift observed in $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures. Consistent with this conclusion, a previous ARPES study on Bi_2Te_3 films grown on Si substrates also reported only a relatively weak thickness-dependent Dirac point shift that is much smaller than 100 meV.⁴⁴ Moreover, in $\text{Bi}_2\text{Te}_3/\text{TiN}$, the pronounced downward shift of the Dirac point from 12 to 24 QL closely resembles the change in T_c observed in transport measurements, whereas such a downward shift is absent in the $\text{Bi}_2\text{Te}_3/\text{InP}$ control samples. Together, these contrasts indicate an additional interface-induced contribution in the $\text{Bi}_2\text{Te}_3/\text{TiN}$ system and suggest that intrinsic bulk defects alone cannot account for the ARPES trends, making interfacial charge transfer a plausible explanation.

To better understand the mechanism of our observations, we carried out first-principles calculations for the $\text{Bi}_2\text{Te}_3/\text{BiTe}/\text{TiN}$ heterostructure on a seven-layer $\text{TiN}(111)$ slab (Figure 4a,b). The charge-density difference indicates that charge transfer is dominated by the $\text{BiTe}-\text{TiN}$ interface, with a smaller contribution at the $\text{Bi}_2\text{Te}_3-\text{BiTe}$ interface. At the $\text{Bi}_2\text{Te}_3-\text{BiTe}$ interface (Figure 4c), pronounced electron-density overlap between Te (Bi_2Te_3) and Bi/Te (BiTe) reveals covalent-like electron sharing that establishes an interfacial charge transfer pathway. Electron accumulation within Bi sites away from this interface further suggests donation from the Bi_2Te_3 side and/or intralayer rearrangement in BiTe . These features are consistent with the planar-averaged charge-density difference along the z direction (right panel of Figure 4c). In contrast, the $\text{BiTe}-\text{TiN}$ interface (Figure 4d) exhibits strong charge rearrangement characterized by electron depletion near Bi and accumulation at N, evidencing direct electron transfer from Bi to the more electronegative N. The corresponding planar-averaged profile shows a negative peak at Bi and a characteristic double positive peak at N, arising from the two-dimensional averaging across distinct charge regions. Electron localization function analysis (Figure 4e) points to predominantly ionic Bi–N bonding, consistent with the charge-density difference results. The states near E_F are dominated by the Ti 3d character, and the interfacial Ti layers

exhibit an enhanced $N(E_F)$ relative to pristine TiN (Figure 4f), consistent with interfacial hybridization and charge transfer.

We also considered both surface terminations of TiN (Ti- and N-terminated) as limiting boundary conditions for the $\text{Bi}_2\text{Se}_3-\text{BiSe}-\text{TiN}$ in our calculations (Figure S12). Work function analysis reveals a thermodynamic driving force for electron transfer from the TI side into TiN for both terminations, with a larger magnitude for N termination. Differential charge-density maps show robust, interface-localized charge redistribution in both cases, indicating that interfacial charge transfer is enabled by the work function mismatch and is robust against the choice of surface termination, with the termination primarily modulating its strength. Layer-resolved band projections for the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure identify the TI and TiN contributions near E_F and visualize the interface coupling (Figure S13). To ensure clear interpretability, we employed a minimal commensurate interface model without a BiSe layer that avoids severe band folding while preserving the key features of charge transfer into TiN and the associated enhancement of Ti-derived states near E_F . We note that the Bi_2Se_3 -projected feature near the Γ point appears to have low dispersion (“flat-like”), which likely originates from interface-induced hybridization and may enhance local DOS, potentially relevant to the superconductivity enhancement.

Studies on Upper Critical Fields

In order to investigate the dimensionality of superconductivity in $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures, we measured the magnetic field-dependent resistance with magnetic fields applied along perpendicular and parallel to the ab plane (Figures S15 and S16). The upper critical fields ($H_{c2,\perp}$ and $H_{c2,\parallel}$), defined by the point where the resistance drops to 50% of the normal-state resistance, are shown in Figure 5b,c. When an external magnetic field is applied to a superconductor, the upper critical field H_{c2} is determined by two main effects.^{12,45,46} The orbital effect arises from the pair breaking induced by the coupling between electron momentum and magnetic field, exhibiting a strong temperature dependence that can be described by the Ginzburg–Landau (GL) theory:

$$H_{c2}^{\perp}(T) = \frac{\Phi_0}{2\pi\xi_{\text{GL}}(0)^2} \left(1 - \frac{T}{T_c} \right) \quad (1)$$

$$H_{c2}^{\parallel}(T) = \frac{\sqrt{3}\Phi_0}{\pi\xi_{GL}(0)d} \left(1 - \frac{T}{T_c}\right)^{1/2} \quad (2)$$

where Φ_0 is the flux quantum, d is the superconducting thickness, and $\xi_{GL}(0)$ is the zero-temperature GL coherence length. The Pauli paramagnetic effect results from the electron spin alignment by a magnetic field and establishes a Pauli limit of H_p ($H_p \approx 1.86T_c$). The orbital-limiting effect typically dominates pair breaking at temperatures near T_c , whereas the Pauli paramagnetic mechanism becomes predominant at low temperatures far from T_c . Figure 5b gives the reduced-temperature (T/T_c) dependence of $H_{c2,\perp}$, exhibiting a clear linear behavior in both the pure TiN film and $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures. For the pure TiN ($n = 0$) sample, the $H_{c2,\perp}$ ($T = 0$ K) is 1.3 T. As Bi_2Te_3 increases, $H_{c2,\perp}$ first increases to a maximum of 1.6 T at $n = 12$ and then slightly decreases at $n = 24$, following the same trend observed in T_c enhancement. The inset of Figure 5b exhibits the extracted GL coherence length $\xi_{GL,\perp}$ extracted by fitting the $H_{c2,\perp}$ using eq 1, which decreases from 15.7 nm in pure TiN to a minimum of 14.4 nm at $n = 12$, and then slightly increases at $n = 24$. Notably, in superconductor–normal metal heterostructures, the superconducting proximity effect typically leads to an increase in the coherence length;¹² however, in ultrathin TiN proximitized by Bi_2Te_3 , the coherence length decreases instead, indicative of an unconventional underlying mechanism.

Figure 5c presents the $H_{c2,\parallel}$ for pure TiN and the 12 nm $\text{Bi}_2\text{Te}_3/4$ nm TiN heterostructure with the highest T_c . According to the GL theory, 2D superconductors exhibit a square-root temperature dependence $H_{c2,\parallel} \propto \sqrt{(1 - T/T_c)}$, while strongly coupled 3D-like superconductors follow a linear behavior of $[H_{c2,\parallel} \propto (1 - T/T_c)]$.⁴⁷ As shown in Figure 5c, the $H_{c2,\parallel}$ for both pure TiN and $\text{Bi}_2\text{Te}_3/\text{TiN}$ is well-described by a linear 3D dependence near T_c , then crossing over to a 2D square-root dependence at lower temperatures. This 3D-to-2D crossover behavior highly resembles the observations in quasi-two-dimensional Nb/Ge composites.⁴⁷ The dimensional crossover is also corroborated by the fitting models with an unfixed exponent in Figure S17b, which gives a fitting exponent of 0.78 lying between 1 and 0.5. Using the 2D model at low temperatures, we extrapolated $H_{c2,\parallel}$ ($T = 0$ K) to be 7.8 T for pure TiN and 8.8 T for 12 nm $\text{Bi}_2\text{Te}_3/\text{TiN}$. In 2D superconductors under in-plane magnetic fields, the orbital pair breaking becomes less effective and the Pauli paramagnetic effect dominates as the temperature decreases. We calculated the Pauli limit of the upper critical field to be 8.0 T for pure TiN and 9.1 T for the 12 nm $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure, highly consistent with the values derived from the 2D GL theory, confirming the Pauli-dominated pair breaking and 2D superconductivity at low temperatures.

DISCUSSION

Based on the data presented in Figure 2a,b, we observed a clear thickness-dependent T_c enhancement in TI/TiN heterostructures. One might attribute the enhanced superconductivity to strain effects, which is more pronounced in ultrathin TiN than in thick TiN. However, in our heterostructures, the TI layer is grown via van der Waals epitaxy, which is believed not to introduce substantial lattice strain into the underlying TiN layer. HRXRD measurements (Figure S2a,b) show that the $\text{TiN}(111)$ peak position remains unchanged within experimental resolution after the growth of Bi_2Se_3 on 70 nm TiN

and Bi_2Te_3 on 4 nm TiN, indicating negligible out-of-plane lattice distortion. Consistently, reciprocal space mapping (Figure S2c) yields layer spacings of $d(111) \approx 2.450$ Å and $d(11\bar{2}) \approx 1.728$ Å, in excellent agreement with reported values for bare TiN.³⁸ Furthermore, STEM-HAADF analysis (Figures S2d and S3d–f) reveals that the in-plane lattice parameter a in both $\text{Bi}_2\text{Te}_3/4$ nm TiN and $\text{Bi}_2\text{Se}_3/4$ nm TiN heterostructures matches bare 4 nm TiN with noise level deviations <1%. Together, these results indicate that strain effects are negligible in both heterostructures. Moreover, the T_c enhancement is absent in both $\text{Se}/4$ nm TiN and $\text{Bi}_2\text{Te}_3/4$ nm TiN with a deteriorated interface, unambiguously confirming that the superconductivity enhancement is not related to strain effects. To summarize the observations in Figure 2, the superconductivity enhancement is exclusively observed at atomically sharp and well-defined interfaces formed by an ultrathin TiN layer and a TI layer incorporating a BiTe (BiSe) bilayer. These results are in stark contrast to the proximity coupling in a superconductor–normal metal interface, indicative of an underlying mechanism distinct from the conventional superconducting proximity effect.^{40,41} Figure 5a shows a clear correlation between the thickness-dependent shift of the Dirac point and the T_c enhancement, consistent with interfacial charge transfer as a likely contributor to the superconductivity enhancement. Comparative analysis of the differential charge maps at Bi_2Te_3 –BiTe and BiTe–TiN interfaces in Figure 4 reveals two complementary transfer mechanisms: electro-negativity-driven, directional injection at BiTe (BiSe)–TiN and covalent-like electron sharing at $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)$ –BiTe (BiSe). Their synergy forms a continuous transfer channel, with the BiTe (BiSe) layer serving as the critical bridge that mediates efficient electron flow from $\text{Bi}_2\text{Te}_3(\text{Bi}_2\text{Se}_3)$ to TiN.

A recent ARPES study indicates that TiN exhibits a sizable Coulomb interaction ($U \sim 6.2$ – 8.5 eV),³⁸ placing it closer to a strongly correlated regime rather than a weak-coupling superconductor. In this context, we do not interpret the interfacial charge transfer as a carrier doping picture. Instead, the charge redistribution at the $\text{Bi}_2\text{Te}_3/\text{TiN}$ interface probably induces a local electronic reconstruction in the interfacial TiN layers by modifying electronic screening, the effective Coulomb interaction U_{eff} and the local bandwidth W . Such changes renormalize the ratio U_{eff}/W , which can shift the system toward a more favorable correlation regime for pairing.

CONCLUSIONS

In summary, we fabricated $\text{Bi}_2\text{Te}_3/\text{TiN}$ and $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures and observed interface-enhanced superconductivity, which occurs exclusively at atomically sharp TI/TiN interfaces featuring an intercalated BiTe (BiSe) bilayer. The ARPES measurements are consistent with interfacial charge transfer, as a likely contributor to the observed superconductivity enhancement. First-principles calculations further elucidate the detailed electron transfer pathways in which the interfacial BiTe bilayer plays a critical role. This work shows that the TI/TiN heterostructure is a promising and tunable TI-SC hybrid system, as the TiN layer eliminates the need for stringent *in situ* growth and maintains sharp superconducting transitions even at ultrathin thicknesses. More importantly, our results offer an effective strategy to manipulate superconductivity in TI-SC heterostructures via interface engineering, i.e., enhancing it with a BiTe interfacial layer or suppressing it by deliberately controlled interface quality, facilitating the search for novel topological superconducting

states. Lastly, our work unveils a new TI-SC family composed of nitride superconductors and topological materials, as a promising candidate for interface-enhanced superconductivity, providing an opportunity for exploring this effect in higher- T_c materials such as NbN or NbTiN ($T_c > 15$ K).

METHODS

Film Growth

Epitaxial TiN(111) films were grown on $5 \times 5 \text{ mm}^2$ α -Al₂O₃(0001) substrates by a homemade magnetron sputtering epitaxy system, the details of which were reported in previous reports.^{37,38} The films in Figure 2b were grown on $5 \times 5 \text{ mm}^2$ KTaO₃(111) substrates. Subsequently, the as-grown TiN films were taken out and transferred into an MBE chamber and annealed at 700 °C under ultrahigh vacuum (UHV) for 12 h to remove surface contaminants. High-purity Bi (99.997%), Se (99.999%), and Te (99.9999%) were evaporated from the Knudsen effusion cells. The TI growth followed the protocols described in our previous report.³⁶ The growth rate was set as ~ 1 QL/min for the TI films, which was calibrated in situ by a quartz crystal microbalance and X-ray reflectivity. Notably, we observed a clear T_c enhancement in Bi₂Se₃/TiN heterostructures grown on both KTaO₃ (Figure 2b) and Al₂O₃ (Figure 2f) substrates.

Crystal Structure and Surface Analysis

Low-energy electron diffraction images were collected *in situ* after sample growth. The crystal structure was characterized by high-resolution X-ray diffraction (HRXRD, Bruker D8 Discovery) measurements with Cu K α radiation ($\lambda = 1.5406$ Å) as the X-ray source. Symmetrical θ – 2θ measurements, rocking curves, and X-ray reflectivity were performed with a 40 kV working voltage and a 40 mA current. The surface morphology was examined by a Bruker Dimension Icon AFM with a Nanoscope V controller (Digital Instruments, USA). All topographic images were acquired in scanAsyst mode with silicon cantilevers. The scanning transmission electron microscopy (STEM) samples were prepared by using focused ion beam (FIB) milling. High-angle annular dark-field scanning transmission (HAADF-STEM) and energy-dispersive X-ray (EDX) spectroscopy measurements were conducted on a Themis Z double spherical aberration-corrected transmission electron microscope with 300 kV.

Transport Measurements

All of the transport measurements were carried out in a Quantum Design physical property measurement system (PPMS) with a base temperature of 1.8 K and a magnetic field up to 9 T. Electrical electrodes were made by manually pressing four indium wires on each sample.

Angle-Resolved Photoemission Spectroscopy Measurements

The electronic structures of Bi₂Te₃/TiN heterostructures were measured at 7.3 K in a homemade angle-resolved photoemission spectroscopy (ARPES) system with a base pressure of $\sim 5 \times 10^{-11}$ mbar. The photoelectrons were excited using a 177 nm deep-ultraviolet laser ($h\nu = 6.997$ eV) with a total energy resolution of better than 1 meV. The Fermi level was determined by the measurement of a polycrystalline gold sample in electrical contact with itself.

First-Principles Calculations

All the calculations were carried out using the Vienna ab initio simulation package (VASP)^{48,49} with the projector-augmented wave (PAW) method. Exchange and correlation effects were treated within the generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) functional.⁵⁰ The Brillouin zone was sampled with a $5 \times 5 \times 1$ gamma-centered Monkhorst–Pack k -point grid. A plane-wave cutoff energy of 500 eV was applied. All atomic positions were fully relaxed until the energy and forces converged to 10^{-5} eV and 0.01 eV/Å, respectively. A vacuum layer of 20 Å was added along the z -direction to avoid interactions between

periodic images. To achieve lattice matching with BiTe, we expanded TiN into a $3\sqrt{3} \times 5 \times 1$ supercell and used its lattice parameters as the reference. Accordingly, BiTe was expanded into a $\sqrt{5} \times \sqrt{5} \times 1$ supercell, and Bi₂Te₃ into a $2\sqrt{3} \times 3 \times 1$ supercell. This configuration introduced no strain to the TiN layer, while the strain in both BiTe and Bi₂Te₃ was kept below 5%, ensuring accuracy of the calculations. The differential charge was defined as $\Delta\rho = \rho(\text{all}) - \rho(\text{TiN}) - \rho(\text{BiTe}) - \rho(\text{Bi}_2\text{Te}_3)$. We also used $\Delta\rho = \rho(\text{all}) - \rho(\text{TiN}) - \rho(\text{BiTe/Bi}_2\text{Te}_3)$ as an alternative approach for verification. The results showed no substantial difference (Figure S14), confirming the robustness of our calculations.

ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c20630>.

HRXRD characterizations, LEED pattern, STEM images and analysis, AFM images, ARPES data, DFT calculations, and transport property data (PDF)

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Author Contributions

[#]R.X., B.H., M.G., and Shenjin Z. contributed equally to this work. X.Y. and Y.C. conceived the experiments. R.X. and X.Y. grew the heterostructures with the help of H.T.Y. and S.O. R.Z., J.B., Shunda Z., and Y.C. grew the TiN films. M.G. performed the STEM measurements and analyzed the data with R.X. R.X., Shenjin Z., F.Z., and S.X. performed the ARPES measurements. R.X. and X.Y. performed the transport measurements. B.H. and T.Z. performed all the calculations. R.X., B.H., T.Z., and X.Y. analyzed the results and wrote the manuscript with contributions from all authors.

Notes

The authors declare no competing financial interest.

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Supplementary Information

Interface-Enhanced Superconductivity in Ultrathin TiN Proximitized by Topological Insulators

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1. High resolution X-ray diffraction characterizations

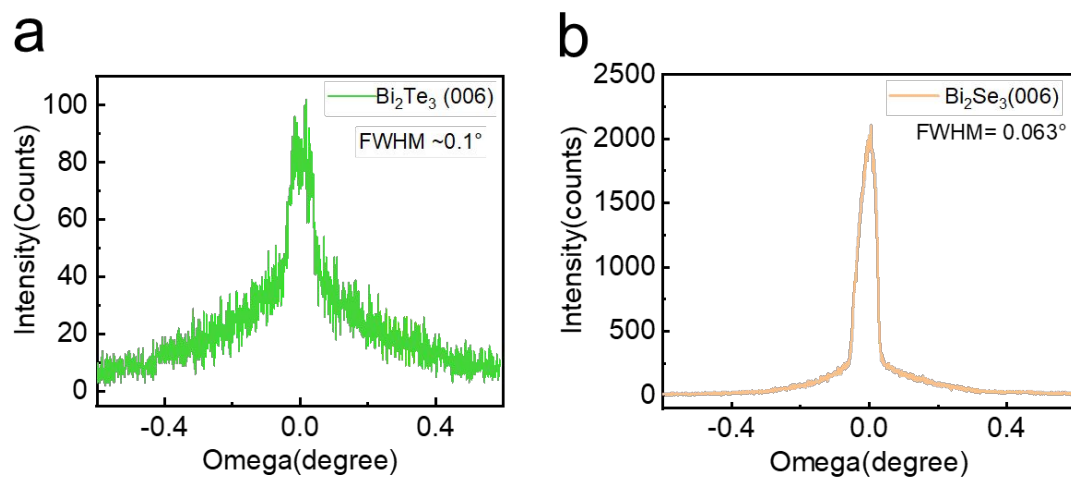


Figure S1. Rocking curves of $\text{Bi}_2\text{Te}_3/\text{TiN}$ and $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructures. (a) 12nm $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure and (b) 50nm $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure. The thickness of the $\text{TiN}(111)$ layer is around 4nm.

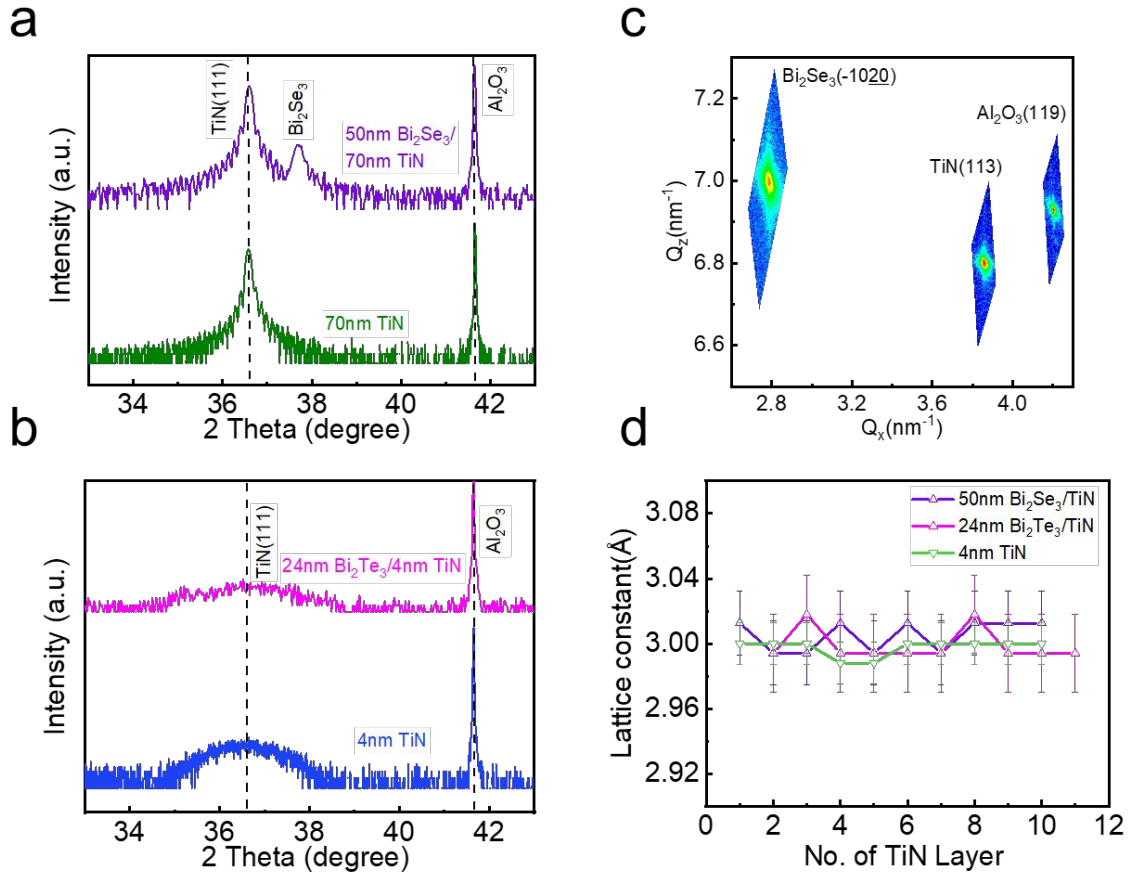


Figure S2. (a, b) The 2 Theta scans around the $\text{TiN}(111)$ diffraction of 70 nm, 4 nm TiN and their TI-SC heterostructures. (c) The reciprocal space mapping (RSM) pattern of 50nm Bi_2Se_3 /70nm TiN on Al_2O_3 substrate. (d) Variation of the in-plane lattice parameter a of 4 nm TiN , 24nm Bi_2Te_3 /4nm TiN , 50nm Bi_2Se_3 /4nm TiN , extracted from their HAADF image (Figure S3d-f). The data were obtained from measurements of atoms located within the white box.

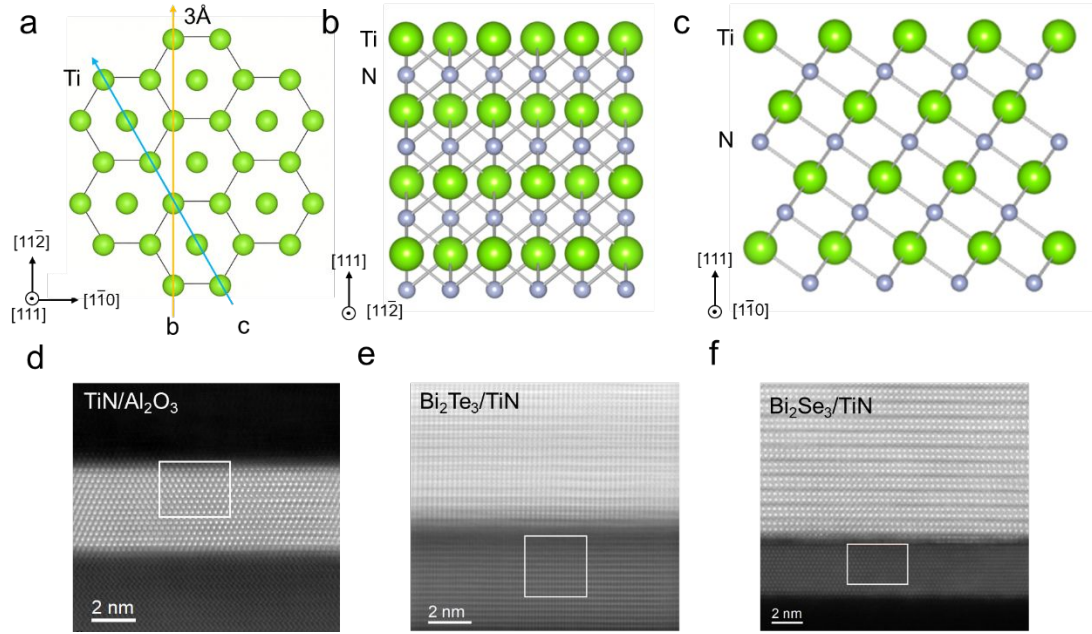


Figure S3. (a) Alternate graphene-like Ti layer, whose lattice parameter a is 3.0 Å. (b, c) b and c present side views of a, illustrating the two orientations shown in Figures 1c and 1e, respectively. In STEM imaging, when the Ti atoms are arranged as b, twice the interatomic distance ($2d$) corresponds to the lattice parameter a . Conversely, when the Ti atoms are arranged as c, the lattice constant a is $(\frac{2}{\sqrt{3}} d)$. (d-f) STEM HAADF images of 4nm TiN, 24nm Bi₂Te₃/4nm TiN and 50nm Bi₂Se₃/4nm TiN, respectively.

2. Low energy electron diffraction patterns

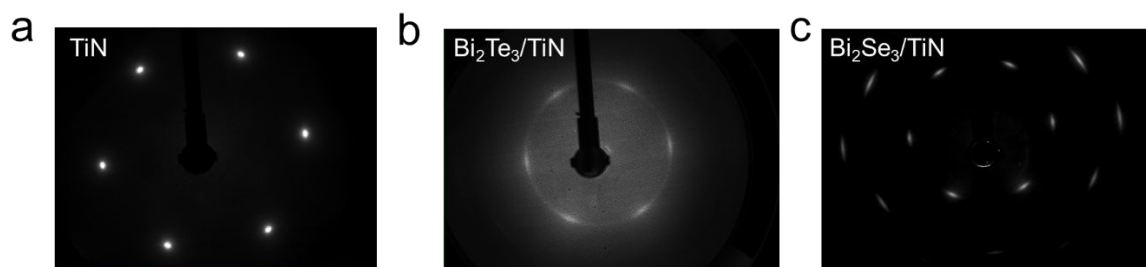


Figure S4. Low energy electron diffraction patterns of (a) TiN film, (b) Bi₂Te₃/TiN heterostructure and (c) Bi₂Se₃/TiN heterostructure. The TiN thickness of all samples is around 4 nm.

3. Scanning transmission electron microscopy characterizations

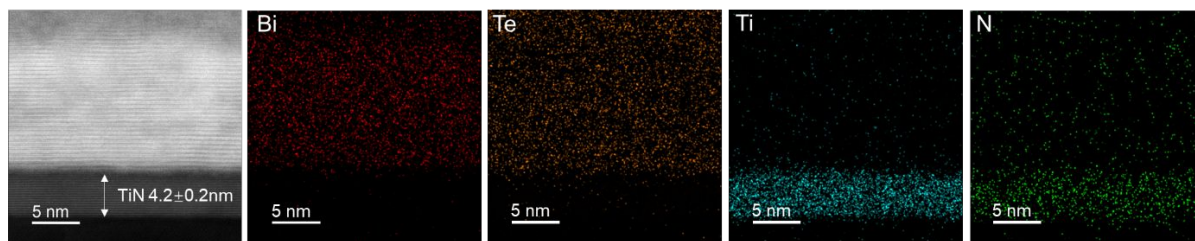


Figure S5. STEM-HAADF images of the $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure, combined with the EDX mapping results for Bi, Te, Ti and N. The EDX mapping of $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure shows a uniform composition, confirming smooth and sharp interface.

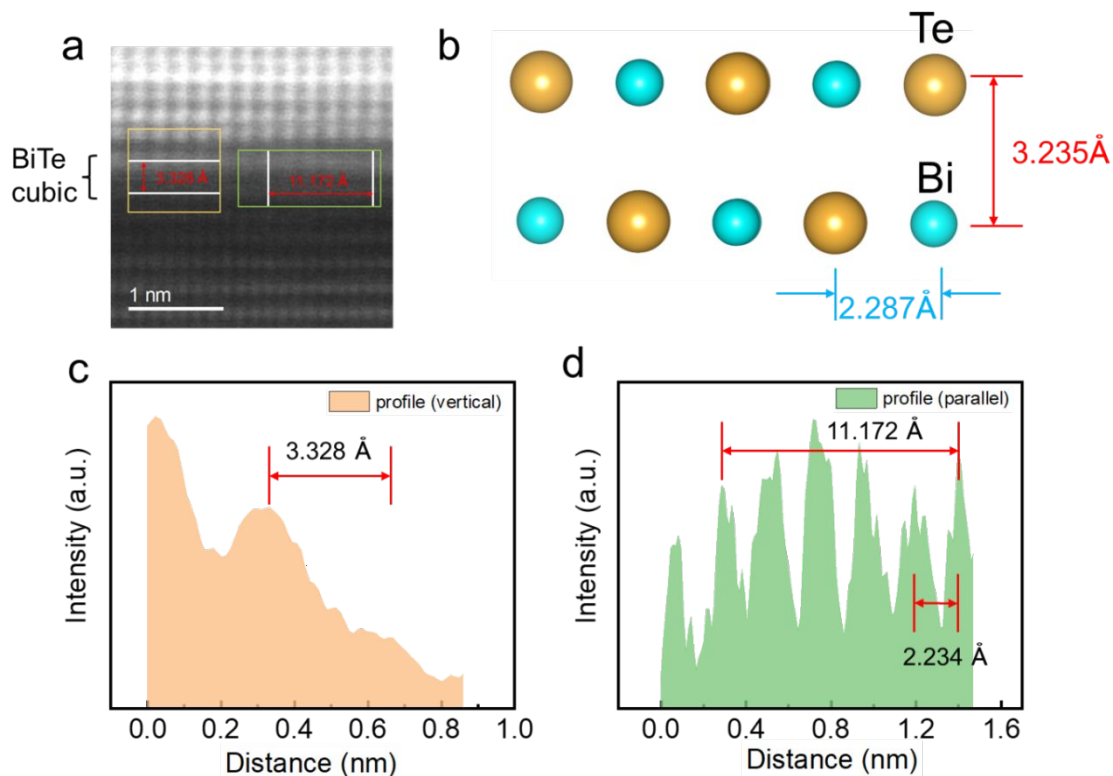


Figure S6. STEM Evidence for the BiTe cubic structure. (a) STEM-HAADF image of the $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure. (b) A schematic illustration of the cubic BiTe bilayer structure. (c-d) Measurements of the atomic distances, corresponding to the yellow and green boxes.

Figure S6c shows an obviously larger interlayer distance than Bi_2Te_3 . The measured out-of-plane atomic distance is 3.328 Å, very close to the theoretical value of 3.235 Å for BiTe and very different from the theoretical value of 2.033 Å for Bi_2Te_3 . The measured in-plane atomic distance is 2.234 Å, very close to the theoretical value of 2.287 Å for BiTe. These results confirm that the bilayer is BiTe cubic structure.

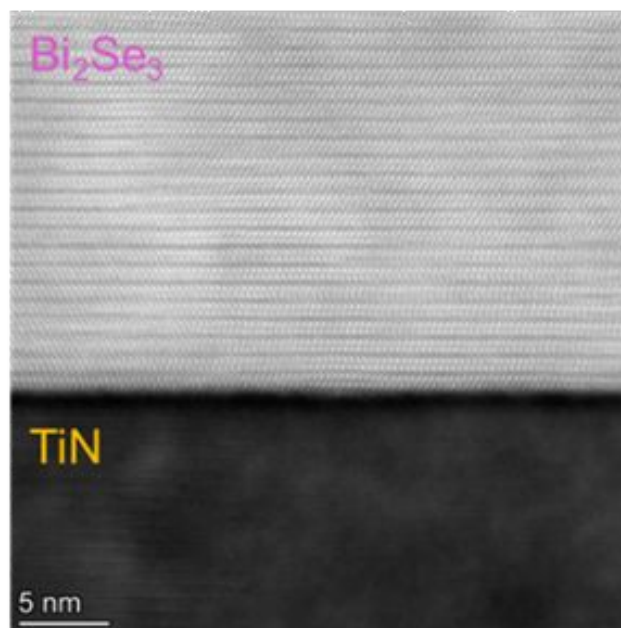


Figure S7. STEM-HAADF image of the Bi₂Se₃/70nm TiN without BiSe layer intercalation.

4. AFM morphology of pure TiN film and Bi₂Te₃/TiN heterostructure

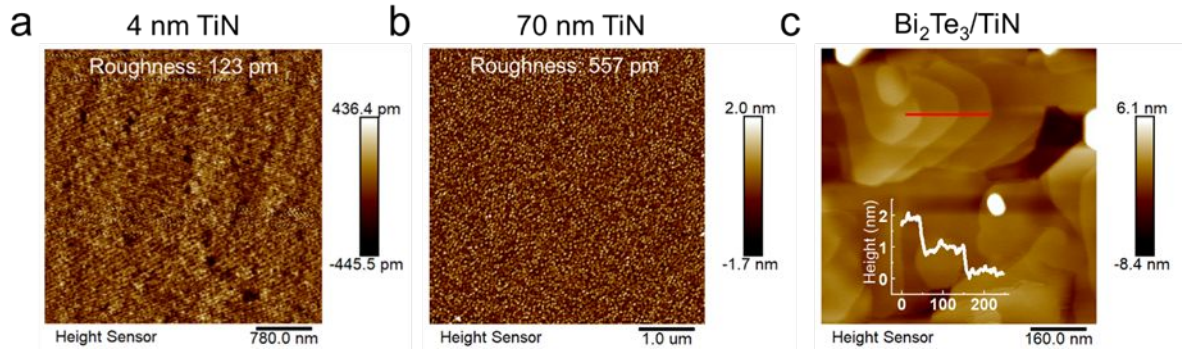


Figure S8. AFM images of (a) pure 4nm-thick TiN(111) film, (b) 70nm-thick TiN (111) films and (c) 12nm Bi₂Te₃/4nm TiN heterostructure.

The surface roughness of the TiN film is around 123 pm, which is even better than commercial Al₂O₃ substrate¹ (146 pm), suggesting the obvious advantage of TiN for TI-SC heterostructures. A large flat terrace is observed in the Bi₂Te₃/TiN heterostructure, suggesting that the well-matched structural symmetry provided by the TiN surface. The inset image shows the height profile of the terraces indicating that the step height of about 1nm, corresponding to 1 QL of Bi₂Te₃.

5. Angle-Resolved Photoemission Spectroscopy Measurements.

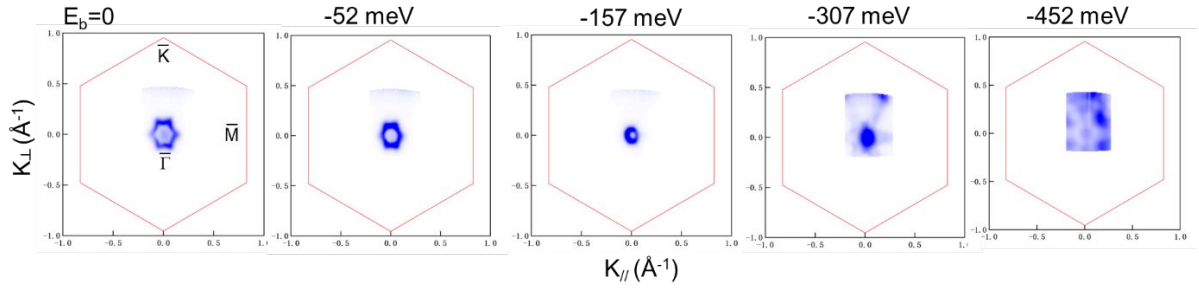


Figure S9. Fermi surface mappings of a 50nm Bi_2Te_3 /4nm TiN heterostructure obtained using a 177 nm deep-ultraviolet laser ($h\nu = 6.997$ eV).

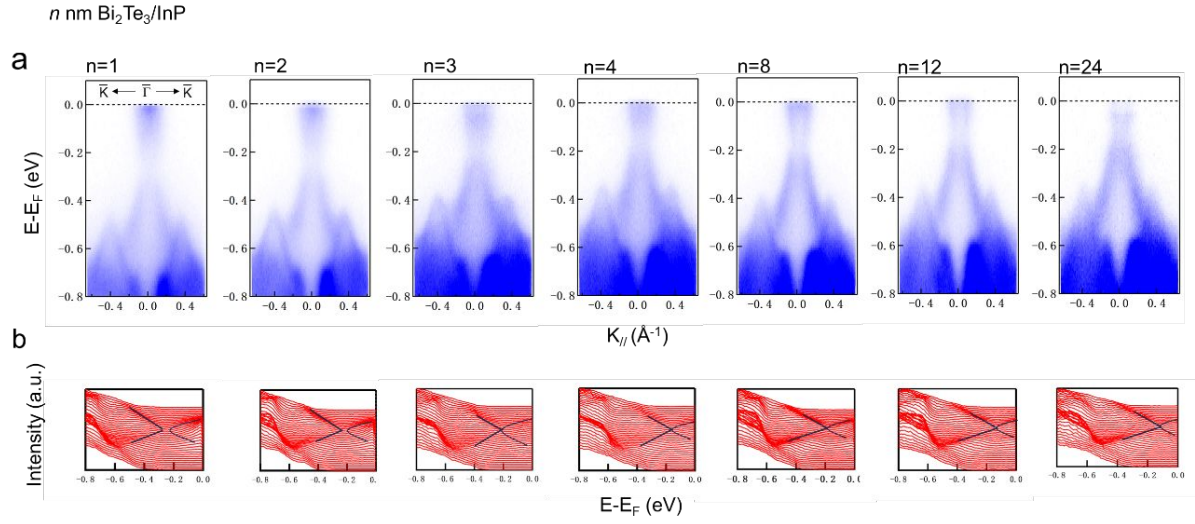


Figure S10: ARPES band spectra of n nm $\text{Bi}_2\text{Te}_3/\text{InP}$ substrates. (a) ARPES results with $n=1, 2, 3, 4, 8, 12, 24$. (b) Corresponding energy distribution curves of ARPES results in (a).

All spectra were collected along the $\bar{\Gamma}-\bar{K}$ direction. The photoelectrons were excited using a He lamp ($h_\nu=21.2$ eV)

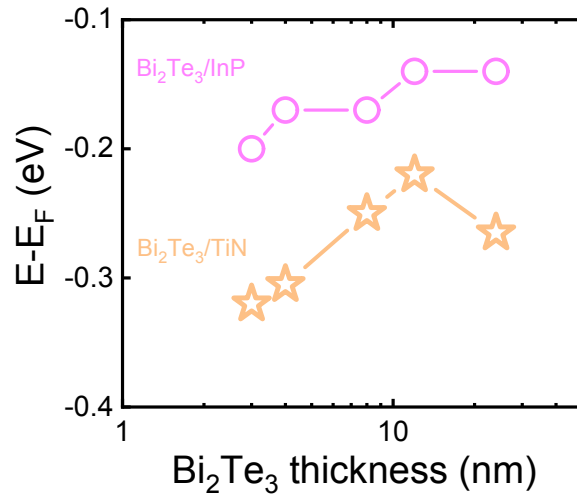


Figure S11 Thickness dependence of the Dirac point ($\text{Bi}_2\text{Te}_3 \geq 3$ nm) in $\text{Bi}_2\text{Te}_3/\text{InP}$ and $\text{Bi}_2\text{Te}_3/\text{TiN}$.

6. First-principles calculations of TiN and heterostructures

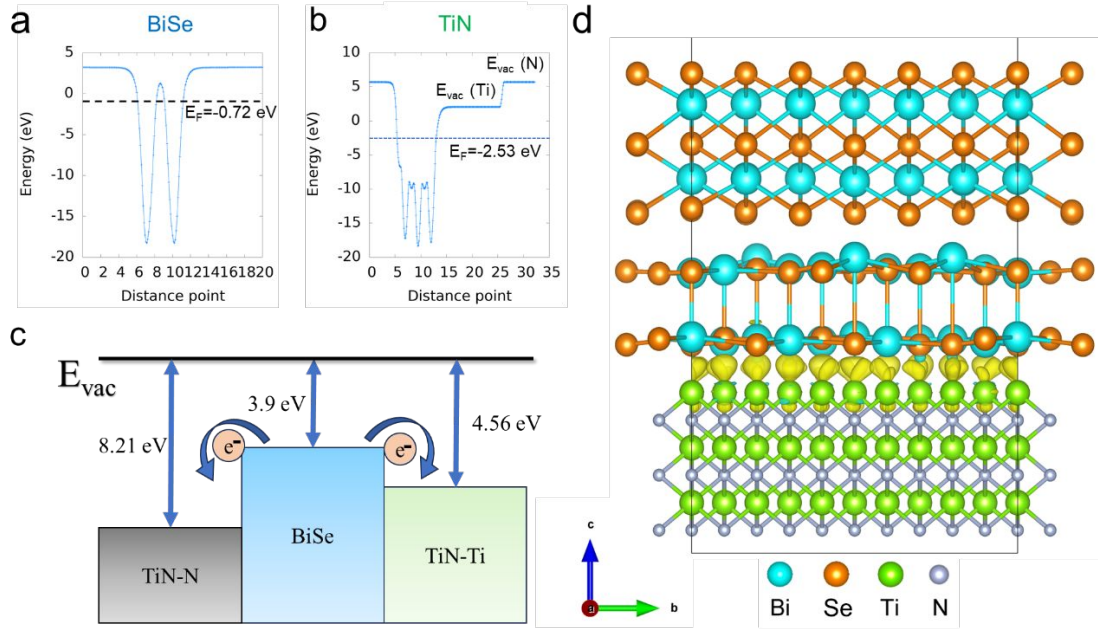


Figure S12 (a) Electrostatic potential profile and extracted work function Φ of the BiSe slab; the vacuum level E_{vac} is obtained from the plateau of the planar-averaged potential in the vacuum region, and $\Phi = E_{vac} - E_F$ (fermi level). (b) Same analysis for TiN slabs with two surface terminations (N-terminated and Ti-terminated). The Fermi level E_F is indicated by the dashed line. The extracted work functions are $\Phi_{TiN(N-term)} \approx 8.21$ eV, $\Phi_{TiN(Ti-term)} \approx 4.56$ eV and $\Phi_{BiSe} \approx 3.9$ eV (c) Schematic energy-level alignment upon contact (Fermi-level equilibration) based on the calculated work functions. The work-function mismatch provides a thermodynamic driving force for electron transfer from BiSe to TiN, with a stronger driving force for the N-terminated TiN surface. (d) 3D differential charge density for the relaxed BiSe/TiN heterostructure. Regions of electron accumulation (depletion) are shown in yellow(blue) at an isovalue of 1-3 $e/\text{\AA}$, indicating charge redistribution localized at the interface.

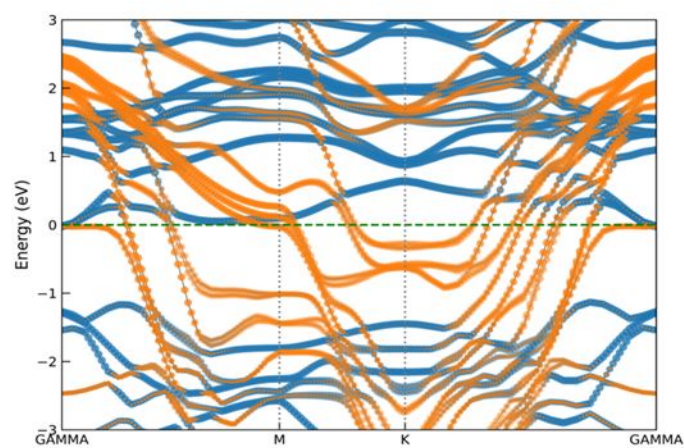


Figure S13 The band diagram of the $\text{Bi}_2\text{Se}_3/\text{TiN}$ heterostructure calculated using a slab model consisting of 1 QL Bi_2Se_3 on a seven-layer $\text{TiN}(111)$ substrate (with a vacuum region of $20 \text{ \AA}$). The Orange and blue bands represent the electronic bands of Bi_2Se_3 and TiN , respectively.

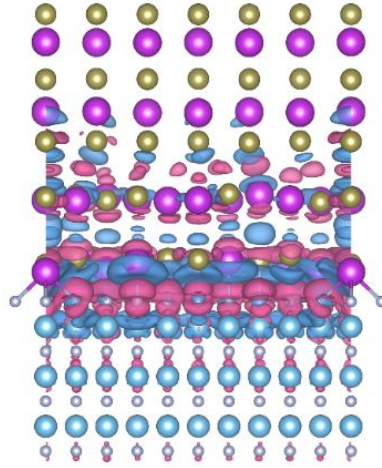


Figure S14. Differential charge density distributions at isosurface level = $0.003 \text{ e}/\text{\AA}$, $\Delta\rho = \rho(\text{all}) - \rho(\text{TiN}) - \rho(\text{BiTe}/\text{Bi}_2\text{Te}_3)$.

7. The transport measurement results of pure TiN and $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructures

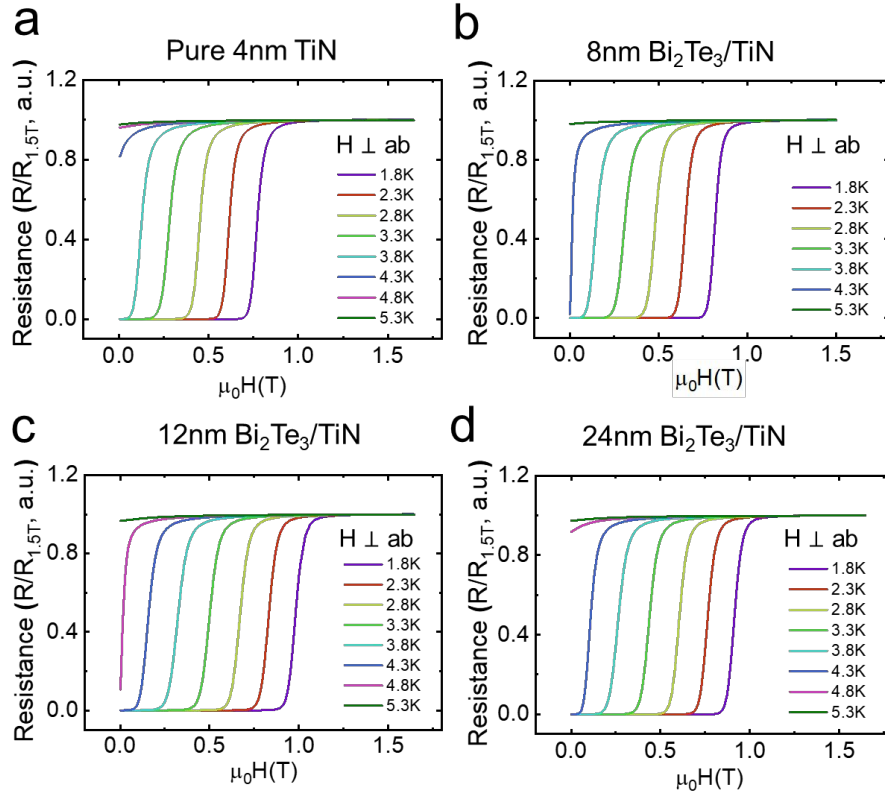


Figure S15. Magnetic field-dependent longitudinal resistance of the pure TiN and $\text{Bi}_2\text{Te}_3/\text{TiN}$

heterostructure with magnetic field applied perpendicular to the ab plane.

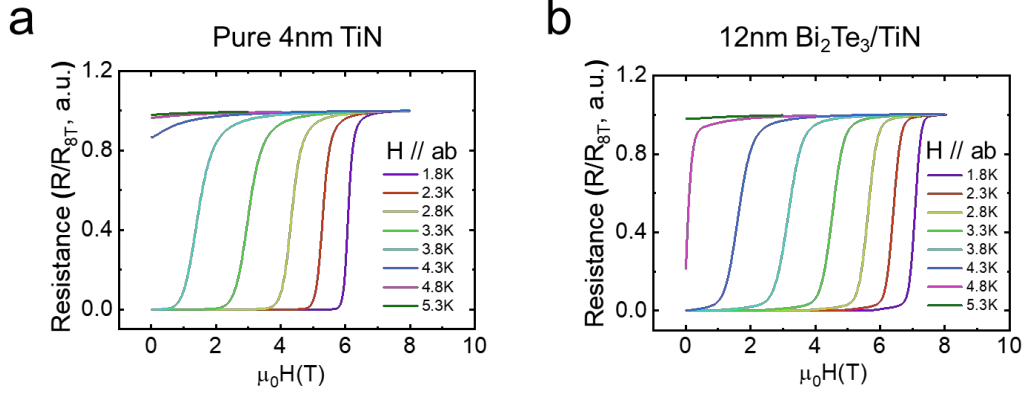


Figure S16. Magnetic field-dependent longitudinal resistance of the pure TiN and 12nm $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure with magnetic field applied parallel to the ab plane.

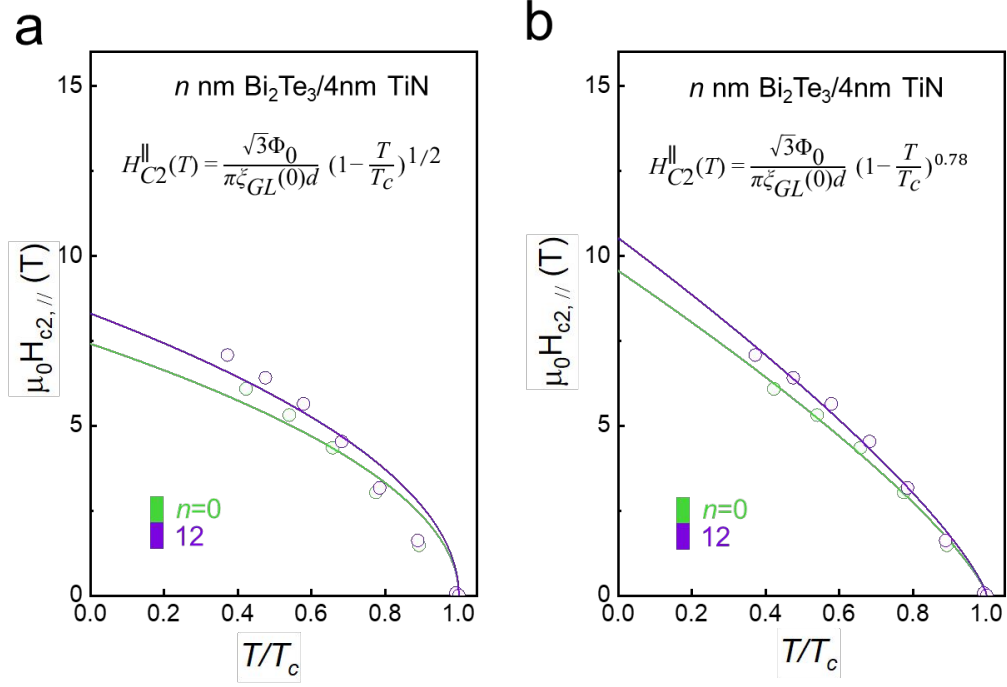


Figure S17. The fitting models of the upper magnetic fields ($H_{c2,||}$) for 4nm TiN film and 12nm $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure. (a) The data is fitted by the Ginzburg-Landau theory and the fitting exponent is 0.5. (b) The fitting exponent is 0.78.

Figure S17 presents the fitting models for the upper critical field $H_{c2,||}$ of pure 4 nm TiN film and 12nm $\text{Bi}_2\text{Te}_3/\text{TiN}$ heterostructure. The fitting model with a $(1 - T/T_c)^{1/2}$ dependence clearly deviates from the experimental data, while the fitting model using $(1 - T/T_c)^{0.78}$ in (b) matches the experimental results closely.

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Resilient Growth of a Highly Crystalline Topological Insulator–Superconductor Heterostructure Enabled

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<https://doi.org/10.1021/acsami.4c05656>.